MILITARY SPECIFICATION

POWER CONTROLLER, SOLID STATE, GENERAL SPECIFICATION FOR

This military specification has been prepared by the Naval Air Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-P-81653. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers the general requirements for the design, manufacture and test of solid state power controllers (see 6.1) of both dc and ac ratings for use in electric power systems.

1.2 Classification. Power controllers shall be classified as specified herein:

1.2.1 Enclosure design. The enclosure design is identified by a single digit as follows:

Type	Enclosure
1	Printed circuit board mounted
2	Enclosed
3	Sealed (other than hermetically)
4	Hermetically sealed

1.2.2 Military part number. The military part number shall consist of the letter "M," the basic number of the specification sheet and an assigned dash number as shown in the following:

Example: M81653 / 00 - 000

- 000 — Dash number
- 00 — Specification sheet number
- M81653 — Military designator

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6110

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2. APPLICABLE DOCUMENTS

2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-T-60	Tape, Pressure Sensitive, Adhesive, Waterproof for Packaging
PPP-T-76	Tape, Pressure Sensitive, Adhesive, Paper, Water Resistant (for Carton Sealing)
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Box Setup
TT-E-527	Enamel, Alkyd, Lustreless
ZZ-R-765	Rubber, Silicone; Low and High Temperature and Tear Resistant

Military

MIL-I-10	Insulating Materials, Electrical, Ceramic, Class L
MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-116	Preservation, Methods of
MIL-P-997	Plastic Material, Laminated, Thermosetting, Electrical Insulation: Sheets, Glass Cloth, Silicone Resin
MIL-C-5015	Connectors, Electric, "AN" Type, General Specification for
MIL-S-12883	Sockets and Accessories, for Plug-In Electric Components, General Specification for

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-15037	Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin
MIL-P-15047	Plastic Material, Laminated, Thermosetting, Sheets, Nylon Fabric Base, Phenolic-Resin
MIL-P-18307	Nomenclature and Identification for Electronic Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-C-28748	Connectors, Electrical, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for
MIL-M-38510	Microcircuits, General Specification for
MIL-G-45204	Gold Plating, Electrodeposited
MIL-C-45662	Calibration System Requirements

STANDARDS

Federal

FED-STD-H28	Screw Thread Standards for Federal Services
FED-STD-595	Color Requirements for Individual Color Chips (3 x 5 Supplements)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment

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STANDARDS (Continued)

Military (Continued)

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Preparation and Submission of Data for Approval of Nonstandard Parts
MIL-STD-1285	Marking of Electrical and Electronic Parts
MIL-STD-1549	Common Termination System for Electrical and Electronic Parts
MS27726	Integrated Wire Termination System for Use on Electrical Components
MIL-HDBK-217	Reliability Prediction of Electromagnetic Equipment

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern. (If a specific requirement specified herein is not required for an item, it shall be so indicated on the specification sheet (e.g., Shock - N/A).)

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3.2 Qualification. Controllers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.4).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the controllers to meet the performance requirements of this specification. Materials used shall be self-extinguishing, and shall not support combustion, give off noxious gases in harmful quantities, give off gases in quantities sufficient to cause explosion of sealed enclosures, cause contamination to any part of the controller, or form current-carrying tracks when subjected to any of the tests specified herein. Unless otherwise specified in the specification sheet, the selection of material shall be such as to provide a shelf-life of at least ten years without affecting the operation of the controller. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.3.1 Fungus-resistant. Materials used in the construction of power controllers shall be fungus inert (Requirement 4 of MIL-STD-454).

3.3.2 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.4 Design and construction. Power controllers shall be of the design, construction, weight, and physical dimensions specified in the specification sheet. Controllers shall be so designed to insure proper operation in any attitude when mounted to a suitable heat sink. The construction of the controllers shall preclude mechanical or electrical damage, flaking of the finish, loosening of the terminals, or deterioration of marking when subjected to the test methods in Section 4 of this specification.

3.4.1 Threaded parts. All threaded parts shall be in accordance with FED-STD-H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series may be used only for applications that show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36, which is used in the fine-thread series. Terminal threads shall be class 2A and 2B for external and internal threads, respectively.

3.4.1.1 Engagement of threaded parts. All threaded external parts shall engage by at least three full threads with all required hardware assembled.

3.4.2 Enclosure. The enclosure for Types 2, 3 and 4 shall be electrically isolated from the circuit and shall provide means for grounding. All exterior surfaces of the enclosure, excluding the mounting surface, shall be finished with lusterless black paint (color 37038) in accordance with FED-STD-595 and MIL-F-14072. Type 1 shall provide an electrically isolated heat sink contact area as specified in the specification sheet. The circuit board for Type 1 controllers shall be conformally coated as specified in the specification sheet.

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3.4.3 Hardware. Hardware shall be as specified in the specification sheet.

3.4.4 Mounting. Mounting shall be as specified in the specification sheet.

3.4.4.1 Bracket. Mounting brackets shall be an integral part of the controller enclosure for Types 2, 3 and 4.

3.4.5 External terminations. External terminations shall be as specified in the specification sheet.

3.4.5.1 Solder. Solder-lug terminals shall be designed to accommodate two conductors and shall be capable of being readily soldered. Unless otherwise specified (see 3.1), solder terminals shall not be gold plated.

3.4.5.1.1 Printed circuit. Terminals shall be placed for compatibility with printed-circuit spacing as specified in the specification sheet.

3.4.5.1.2 Solderability. For qualification inspection, controllers shall be tested as specified in 4.8.3 and shall meet the applicable requirements of Method 208 of MIL-STD-202.

3.4.5.1.3 Resistance to soldering heat. Where specified (see 3.1) qualification inspection, controllers shall be tested as specified in 4.8.17, in accordance with Method 210 of MIL-STD-202, and shall subsequently meet the requirements of paragraph 3.13.1.

3.4.5.2 Plug-in (connector). Terminals for plug-in headers shall conform to the dimensions and contact arrangements necessary for proper mating with the applicable terminations conforming to MIL-C-5015, MIL-C-8384 or MIL-S-12883. Pin contacts only shall be used. Unless otherwise specified in the specification sheet, terminals shall be gold plated in accordance with MIL-G-45204, Type II, Class 1 over nickel plate 0.0001 to 0.0003 inch thick.

3.4.5.3 Integrated. Controllers designed for removable and insertable terminations shall be in accordance with MIL-STD-1549.

3.4.5.4 Deleted.

3.4.5.5 Maintainability. The power controller and its components shall meet the requirements of this specification without the need for any routine servicing or maintenance actions. The power controller design shall not include any provision for the disassembly or inspection of the power controller package or its components.

3.5 Conditioning (run-in). When tested as specified in 4.8.2, the controller shall perform satisfactorily during each operation.

3.6 Seal. When tested as specified in 4.8.4, the leakage rate shall not exceed 10^{-6} atm cc/sec for type 4 controllers.

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3.7 Dielectric withstanding voltage. When tested as specified in 4.8.5, there shall be no leakage current in excess of 1.0 milliampere (ma) nor evidence of damage due to arcing (air discharge), flashover (surface discharge), or insulation breakdown (puncture discharge).

3.8 Insulation resistance. When tested as specified in 4.8.6, the insulation resistance shall be greater than 100 megohms.

3.9 Electrical characteristics. When tested as specified in 4.8.7, controllers shall operate with supply voltage variations in accordance with MIL-STD-704. Controllers shall be capable of controlling reactive and resistive loads.

3.10 Performance.

3.10.1 Control signal. When tested as specified in 4.8.7.1, the control signal shall be as specified in the specification sheet.

3.10.2 Turn-on and turn-off time. When tested as specified in 4.8.7.2, the turn-on and turn-off time shall be as specified in the specification sheet.

3.10.3 Load voltage rise and fall time. When tested as specified in 4.8.7.3, the rise and fall time shall be as specified in the specification sheet.

3.10.4 Isolation. The control/power isolation test voltage shall be as specified in the specification sheet. When tested as specified in 4.8.7.4, the controller shall have a minimum isolation resistance of 100 megohms.

3.10.4.1 Isolation between controllers. For multiple controllers packaged on a single printed circuit board, the controllers shall have a minimum isolation resistance of 100 megohms between any two controllers. The isolation test voltage shall be as specified in the specification sheet and tests shall be in accordance with 4.8.7.4.1.

3.10.5 Control and built-in test (BIT) signal levels.

3.10.5.1 Control signal levels. When tested as specified in 4.8.7.5, the control signal levels shall be as specified (see 3.1). Where maximum control signals are specified (see 3.1) the signal shall be applied for 10 minutes without damage to the controller.

3.10.5.2 BIT signal levels. When BIT is specified (see 3.1) the signal levels shall be developed and tested as specified (see 3.1). The signals once developed shall remain constant for sampling period.

3.10.6 Voltage drop. When tested as specified in 4.8.7.6, the voltage drop shall not exceed the value specified in the specification sheet for load current values from no load to 100 percent rated.

3.10.7 Leakage current. When tested as specified in 4.8.7.7, the leakage current shall not exceed the values specified in the specification sheet.

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3.10.8 Power dissipation. When tested as specified in 4.8.7.8, the power dissipation shall not exceed the values specified in the specification sheet.

3.10.9 Overload characteristics.

3.10.9.1 Current limiting. When specified in the specification sheet and tested as specified in 4.8.7.9.1, the output current shall be within the limits specified. At the initiation of the overload condition, the peak let through current (see 6.5.27) shall not exceed the value specified.

3.10.9.2 Trip time.

3.10.9.2.1 Nonrepetitive reset. When tested as specified in 4.8.7.9.2(a), the trip time shall be within the limits specified in the specification sheet.

3.10.9.2.2 Repetitive reset. When tested as specified in 4.8.7.9.2(b), the controller shall reset; the trip time shall be less than the time specified in the specification sheet. The controller shall not be damaged.

3.10.10 Coordination. When tested as specified in 4.8.7.10, only the lower rated power controller shall trip, in the series connected pair of power controllers of rating ratios as specified in the specification sheet (see 3.1).

3.10.11 Rupture capacity. When tested as specified in 4.8.7.11, the controller shall trip and there shall be no damage to the controller. The controller shall be resettable within 10 minutes after each test.

3.10.12 Reset function. When tested as specified in 4.8.7.12, the controller shall not reset when the control signal is switched from ON to OFF for a time duration less than the minimum time specified in the specification sheet and then the ON signal is reapplied. The controller shall reset when the control signal is switched from ON to OFF for greater than the minimum time specified and then the ON signal is reapplied.

3.10.13 Deleted.

3.10.13.1 Deleted.

3.10.13.2 Deleted.

3.10.14 State indication. The power controllers shall provide the means of state indication specified in the specification sheet when tested as specified in 4.8.7.14. State indication shall include the detection of load current above or below a minimum current threshold and the presence or absence of drive to the output power switches as specified in the specification sheet. These state indication means, in conjunction with the control signal, shall be capable of providing feedback on normal controller operation or controller faults as specified in the specification sheet.

3.10.14.1 State indication delay. The response time to a change of state shall not exceed the time specified in the specification sheet.

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3.10.15 Deleted.

3.10.16 Ripple voltage (dc controller). When tested as specified in 4.8.7.16, the ripple voltage (see 6.5.27) generated by the dc controller shall not exceed the value specified in the specification sheet (see 3.1).

3.10.17 Trip-free characteristics. When tested as specified in 4.8.7.17, the controller shall reset, trip-out and stay tripped out for the duration of the test.

3.10.18 Operating voltage transients. When tested as specified in 4.8.7.19(a), the controller shall not be damaged or deviate from the OFF state. When tested as specified in 4.8.7.19(b), the controller shall not be damaged or tripped, but may go to the OFF state for the time specified in the specification sheet.

3.10.19 Transient spike overvoltage. When tested as specified in 4.8.7.20, the controller shall not be damaged, shall stay ON when controlled ON and shall stay OFF when controlled OFF.

3.10.20 Control/status transients. When tested as specified in 4.8.7.22, the controller shall not be damaged. When the control circuit is tested, the controller may momentarily change states but shall not permanently turn on.

3.10.21 Zero voltage turn-on zero current turn-off. When tested as specified in 4.8.7.23, controller turn-on shall occur at zero voltage cross-over within the voltage or time specified (see 3.1), and the controller turn-off shall occur at zero current crossover within the current or time specified (see 3.1). The controller shall turn-on and turn-off at the same voltage slope (turn-off at the opposite half-cycle from turn-on), when specified (see 3.1).

3.10.22 Common mode rejection. When tested as specified in paragraph 4.8.7.25, the controller shall remain in the turned ON or turned OFF state as commanded.

3.10.23 dc offset voltage (ac controllers). When tested as specified in 4.8.7.26, the dc offset voltage shall not exceed the value specified for the load current values specified (see 3.1).

3.10.24 Stabilization time. Controllers shall not pass power to the output during the stabilization time unless commanded to do so.

3.11 Fail-safe. The controllers shall incorporate a "fail-safe" feature in the event the "trip-circuit" fails to perform its function during an over-load condition. When tested as specified in 4.8.8, the fail-safe element shall open the circuit at currents and time specified (see 3.1).

3.12 Environmental characteristics. Unless otherwise specified, the controller shall meet the requirements of paragraph 3.12.1 when subjected to the following tests:

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Temperature shock	4.8.9
Vibration	4.8.11
Acceleration	4.8.12
Salt fog	4.8.13
Temperature altitude	4.8.14
Humidity	4.8.16
Resistance to soldering heat	4.8.17
Shock	4.8.19
Life	4.8.20
Explosive atmosphere	4.8.8
Explosive decompression	4.8.21

3.12.1 Performance criteria. Unless otherwise specified, the controller shall meet the following criteria when subjected to the tests given in 3.12.

Design and construction	3.4
Dielectric withstanding	3.7
Insulation resistance	3.8
Electrical characteristics	3.9

3.13 Terminal strength. When Type 2, 3 and 4 controllers are tested as specified in 4.8.15, there shall be no loosening or breaking of the terminals and there shall be no deformation to the threads of screw terminals, no bending (1/) of a pin or damage to the insulating base of plug-in controllers, nor shall there be any other damage which would adversely affect the normal operation of the controller.

1/ Bending of solder terminals shall not be construed as damage; bending of plug-in terminals shall not be construed as damage, provided they can be reformed in a manner to permit proper mating with the applicable socket.

3.14 Electromagnetic compatibility. When tested as specified in 4.8.18, controllers shall meet the requirements of MIL-STD-461 Class A1.

3.15 Case-to-sink thermal resistance. The thermal resistance shall be as specified in 3.1 when mounted dry to a flat metal plate (0.001 inch per inch) with a surface finish of 63 microinches or less and a mounting torque as specified (see 3.1).

3.16 Marking. Controllers shall be marked in accordance with the requirements of MIL-N-18307. In addition, the following information shall be provided.

(a) Rated control signals (see 3.1).

(b) Rated output voltage and current (see 3.1).

(c) Terminal nomenclature (see 3.1).

(d) Manufacture date code in accordance with MIL-STD-456

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3.17 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.18 Operation at temperature extremes. The controller shall meet the electrical characteristics of 3.11 when tested in accordance with paragraph 4.8.10.

3.19 Life. When power controllers are tested in accordance with paragraph 4.8.20, there shall be no evidence of performance degradation.

3.20 Explosive decompression. The controllers shall not be damaged when tested as specified in 4.8.21.

3.21 Reliability. When operated within the environmental and operational limits specified herein, the controllers shall achieve a reliability, expressed in a mean time between failures (MTBF), as specified in the specification sheet. Failure is defined as the inability to provide all the performance characteristics specified herein. Reliability requirements will be verified by analysis per MIL-HDBK-217.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).

(b) Qualification inspections (see 4.5).

(c) Quality conformance inspection (see 4.7).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the power controllers are in accordance with the applicable referenced specifications prior to such fabrication.

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TABLE I. Materials inspection. Deleted.

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-810.

4.4.1 Power supply. The power supply (see Figure 1) shall have no more than 10 percent regulation at twice the specified load current. A dc power supply shall have no more than 5 percent voltage ripple. An ac power supply shall be within 1 percent of the specified frequency and shall be sinusoidal with a form factor between 0.95 and 1.25.

4.4.1.1 Grounding. The negative side of the dc power supply shall be grounded; one side of the single phase ac power supply shall be grounded, or the neutral of the three-phase ac power supply shall be grounded, as applicable. The grounds of the control circuits in the controller being tested shall be connected to the grounded side of their power supply. The load of the controller shall be connected to the grounded side of the output power supply.

4.4.2 Load conditions during test. Full load shall imply operation by the specified (see 3.1) control parameters with the maximum rated voltage.

4.4.3 Heat removal. Heat removal apparatus shall consist of a heat sink of known characteristics (see 3.15) which can be adjusted and monitored to provide the cooling necessary to prevent the heat sink temperature from exceeding the specified maximum.

4.4.4 Temperature stabilization of test items.

4.4.4.1 Nonoperating. Temperature stabilization shall occur when the case temperature does not differ more than 2.0°C from the ambient temperature.

4.4.4.2 Operating. Temperature stabilization shall occur when the case temperature does not change more than 2.0°C (3.6°F) per hour.

4.4.5 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.4) on sample units selected from a production lot.

4.5.1 Sample size. The number of power controllers to be subjected to qualification inspection shall be as specified in Table II.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in Table II, in the order shown. All sample units shall be subjected to the inspections of Group I. The sample shall then be divided as specified in Table II, for Groups II and III (as applicable) and subjected to the inspections for their particular group.

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4.5.3 Retention of qualification. To retain qualification, the supplier shall forward, at 24-month intervals, to the qualifying activity, a summary of Group A tests, indicating as a minimum the number of lots which passed and the number which failed, and the complete results of Group B tests, including the number and type of any part failure. The summary shall include those tests performed during the 24-month period. If the summary of the test results indicates nonconformance with specification requirements, action shall be taken to remove the failing product from the qualified products list. Failure to submit the summary shall result in loss of qualification for that product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 24-month period that the inspection data indicates failure of the qualified product to meet the requirements of the specification. In the event that no production occurred during the reporting period, a report to that effect shall be submitted.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Group A inspection. Except as specified in 4.7.2.2.3, delivery of products which have passed Group A inspection shall not be delayed pending the results of the Group B and C inspections.

4.6.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all power controllers of the same enclosure, configuration, vibration characteristic, temperature class, shock type, produced under essentially the same conditions within a period not to exceed four weeks, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in Table III, in the order shown.

4.6.1.2.1 Sampling plan. Sampling plan shall be as specified in Table III. Statistical sampling and inspection for Group A-II shall be in accordance with MIL-STD-105 for general inspection level II. Acceptable Quality Level (AQL) shall be as specified in Group A-II Table III. Samples which fail Group A-II may be reworked and resubmitted using the tightening inspection in accordance with MIL-STD-105.

4.6.1.2.2 Disposition of sample units. Sample units which have been subjected to and have passed Group A inspection shall be delivered on the contract or purchase order.

4.6.1.2.2.1 Rejection samples. Rejection samples for Group A-I may be reworked and resubmitted for inspection.

4.6.2 Periodic inspection. Periodic inspections shall consist of Groups B and C inspections.

4.6.2.1 Group B inspections. Group B inspection shall consist of the test(s) specified in Table IV and shall be performed in the order shown.

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TABLE II (4.5.2). Qualification inspection.

Examination or Test	Requirement Paragraph	Method Paragraph	No. of Sample Units to be Tested	No. of Failures Allowed
Group I				
Conditioning (run-in)	3.6	4.8.2		
Visual and mechanical examination:	3.1, 3.4	4.8.1		
External	3.5, 3.17, 3.18,			
Solderability	3.5.5.1.2	4.8		
Seal	3.8	4.8.4		
Dielectric withstanding voltage	3.9	4.8.5		
Insulation resistance	3.10	4.8.6		
Electrical characteristics:	3.11	4.8.7		
Control signal	3.11.1	4.8.7.1		
Turn-off time	3.11.2	4.8.7.2		
Turn-on time	3.11.2	4.8.7.2		
Load voltage rise and fall time	3.11.3	4.8.7.3		
Isolation	3.11.4	4.8.7.4		
Control input	3.11.5	4.8.7.5		
Voltage drop	3.11.6	4.8.7.6		
Leakage current	3.11.7	4.8.7.7		
Power dissipation	3.11.8	4.8.7.8	2	0
Overload characteristics	3.11.9	4.8.7.9		
Coordination	3.11.10	4.8.7.10		
Rupture capacity	3.11.11	4.8.7.11		
Reset signal	3.11.12	4.8.7.12		
Reset time	3.11.13	4.8.7.13		
State indication	3.11.14	4.8.7.14		
Reset immunity	3.11.15	4.8.7.15		
Ripple current	3.11.16	4.8.7.16		
Trip free characteristics	3.11.17	4.8.7.17		
Operating voltage transients	3.11.18	4.8.7.19		
Transient spike overvoltage	3.11.19	4.8.7.20		
Control/input transients	3.11.21	4.8.7.22		
Zero voltage turn-on and zero current turn-off	3.11.22	4.8.7.23		
Common mode rejection	3.11.24	4.8.7.25		
dc offset voltage	3.11.25	4.8.7.26		
Stabilization time	3.11.26			
Fail safe	3.12	4.8.8		
Group II				
Temperature shock	3.13	4.8.9		
Vibration	3.13	4.8.11		
Acceleration	3.13	4.8.12		
Salt fog	3.13	4.8.13		
Temperature altitude	3.13	4.8.14		
Terminal strength	3.13	4.8.15	2	1
Humidity	3.13	4.8.16		
Seal	3.8	4.8.4		
Operational temperature extremes	3.19,	4.8.10		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.17, 3.18	4.8.1		
Group III				
Electromagnetic compatibility	3.15	4.8.18		
Terminal strength	3.14	4.8.13		
Shock	3.13	4.8.19		
Life 1/	3.13	4.8.20		
Seal	3.8	4.8.4		
Electrical characteristics	3.11	4.8.7		
Control signal	3.11.1	4.8.7.1		
Turn-off time	3.11.2	4.8.7.2	2	1
Turn-off time	3.11.2	4.8.7.2		
Control input	3.11.5	4.8.7.3		
Output voltage drop	3.11.6	4.8.7.6		
Power dissipation	3.11.8	4.8.7.8		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.17, 3.18	4.8.1		

1/ Four extra samples (a total of 8) shall be subjected to life test.

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4.6.2.1.1 Sampling plan. The number of sample units specified in Table IV shall be selected every six months from inspection lots which have passed Group A inspection.

4.6.2.1.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for inspection. Resubmitted lots shall be subjected to Groups A and B inspection. Such lots shall be separated from new lots, and shall be clearly identified as reworked lots.

4.6.2.1.3 Disposition of sample units. Sample units which have been subjected to a Group B inspection shall not be delivered on the contract or purchase order.

4.6.2.2 Group C inspection.

4.6.2.2.1 Sampling plan. Every 24 months, controllers shall be subjected to the inspections of Table II, performed in the order shown. The number of failures allowed shall be as specified in Table II.

4.6.2.2.2 Disposition of sample units. Unless otherwise specified, sample units which have been subjected to Group C inspection shall not be delivered on the contract or purchase order.

4.6.2.2.3 Noncompliance. If a sample fails to pass Group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, Group C inspection shall be repeated on additional sample units. Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until Group C reinspection has shown that the corrective action was successful.

4.6.3 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5.

4.7 Deleted.

4.8 Methods of examination and tests. When applicable, testing of power controllers with plug-in terminations shall be performed with the appropriate or specified socket or connector mated to the controller.

4.8.1 Visual and mechanical examination. Power controllers shall be examined to verify that the materials, external design and construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.3, 3.4, 3.16 and 3.17).

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TABLE III. Group A inspection.

Examination or Test	Requirement Paragraph	Method Paragraph	No. of Sample Units to be Inspected
Group I			
Conditioning (run-in)	3.6	4.8.2	
Dielectric withstanding voltage	3.9	4.8.5	100%
Insulation resistance	3.10	4.8.6	Inspection
Electrical characteristics:	3.11	4.8.7	
Turn-off voltage	3.11.1	4.8.7.1	AQL Not Applicable
Turn-on voltage	3.11.1	4.8.7.1	
Turn-off time	3.11.2	4.8.7.2	
Turn-on time	3.11.2	4.8.7.2	
Control input	3.11.5	4.8.7.5	
Voltage drop	3.11.6	4.8.7.6	
Leakage current	3.11.7	4.8.7.7	
Overload characteristics	3.11.9	4.8.7.9	
Coordination	3.11.10	4.8.7.10	
Reset voltage	3.11.12	4.8.7.12	
Reset time	3.11.13	4.8.7.13	
State indication	3.11.14	4.8.7.14	
Zero voltage crossover turn-on and zero current cross-over turn-off	3.11.22	4.8.7.23	
Common mode rejection	3.11.24	4.8.7.25	
dc offset voltage	3.11.25	4.8.7.26	
Group II			100% Inspection
Visual and mechancal examination (external)	3.1, 3.4, 3.5, 3.17, 3.18	4.8.1	AQL Percent Defective Major 1.0% Minor 4.0%

TABLE IV. Group B inspection.

Examination or Test	Requirement Paragraph	Method Paragraph	No. of Sample Units to be Tested	No. of Failures Allowed
Dielectric withstanding voltage	3.9	4.8.5	2	0
Insulation resistance	3.10	4.8.6		
Turn-off voltage	3.11.1	4.8.7.1		
Turn-on voltage	3.11.1	4.8.7.1		
Power dissipation	3.11.8	4.8.7.8		
Operating voltage transient	3.11.19	4.8.7.19		
Temperature shock	3.13	4.8.9		
Seal	3.8	4.8.4		

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TABLE V. Performance under environmental conditions.

Test paragraph	Requirement paragraph																	
	3.10.1	3.10.2	3.10.3	3.10.4	3.10.6	3.10.7	3.10.8	3.10.9	3.10.12	3.10.14	3.10.16	3.10.17	3.10.18	3.10.19	3.10.20	3.10.21	3.10.22	3.10.23	3.10.24
4.8.9	X				X	X			X	X								X	
4.8.10	#	#	#		#	#	#	#	#	#	#	#	#	#	#	#	#	#	#
4.8.11	X				X	X			X	X								X	
4.8.12	X				X	X			X	X								X	
4.8.13	X				X	X			X	X								X	
4.8.14	X								X	X									
4.8.16	#								#	#								#	#
4.8.17	X				X	X			X	X								X	
4.8.19	X				X	X			X	X								X	
4.8.20	#	#	#	#	#		#	#	#	#	#	#	#	#	#	#	#	#	#
4.8.21	X				X	X			X	X								X	
4.8.22	X				X	X			X	X								X	

- Measurement performed during test.
X - Measurement performed after test.

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4.8.2 Conditioning (run-in). Power controllers shall be subjected to 5000 operations at a rate of 1 to 60 operations per second under the following conditions:

(a) The control input shall be energized at the rated turn-on signal (see 3.1).

(b) Unless otherwise specified, the load shall be rated resistive load.

(c) The unit shall be cycled at 50 ±5 percent duty cycle.

4.8.3 Solderability. Controllers shall be tested in accordance with Method 208 of MIL-STD-202. The following shall apply:

(a) All terminations of each part shall be tested.

(b) Depth of immersion in flux and solder - Leads shall be immersed to within 1/16 inch of the seal or case.

4.8.4 Seal (see 3.6).

4.8.4.1 Type 3 controllers. Controllers shall be tested in accordance with Method 112 of MIL-STD-202. The following shall apply:

(a) Test Condition - B.

(b) Measurements after test - Not applicable.

4.8.4.2 Type 4 controllers. Controllers shall be tested in accordance with Method 112 of MIL-STD-202. The following shall apply:

(a) Test Condition C (Procedure III or IV, at the option of the manufacturer). Test Condition B shall be used to test for gross leaks.

(b) Leakage rate - 10^{-6} atm cc/sec.

(c) Reduced pressure of chamber and duration of pressurization (Procedure IV) - Below 5 mm of mercury for 6 hours.

(d) Measurements after test - Not applicable.

4.8.5 Dielectric withstanding voltage (see 3.7). Controllers shall be tested as specified in 4.8.5.1 and 4.8.5.2.

4.8.5.1 At atmospheric pressure. Controllers shall be tested in accordance with Method 301 of MIL-STD-202. The following details shall apply:

(a) Preparation - Not applicable.

(b) Test voltage - As specified in the specification sheet.

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(c) Nature of potential - ac.

(d) Duration - 60 seconds.

(e) Points of application - All terminals shall be shorted together and the test voltage applied from terminals to case.

(f) Leakage current - 10 ma maximum.

(g) Following these tests, controllers shall be examined for evidence of arcing, flashover, insulation breakdown and damage.

(h) Measurements after test - Insulation resistance.

4.8.5.2 At reduced barometric pressure. Controllers specified for operation above 10,000 feet shall be tested in accordance with Method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Normal mounting means.

(b) Test condition - D.

(c) Test requirements at reduced pressure - As specified in 4.8.5.1 except test voltage shall be as specified in the specification sheet.

4.8.6 Insulation resistance. Controllers shall be tested in accordance with Method 302 of MIL-STD-202. The following details shall apply:

(a) Test Condition - A.

(b) Preparations - None.

(c) Points of measurements - The terminals shall be shorted together and measurements taken between enclosure and terminals.

(d) Electrification time - 2 minutes.

(e) Measurement error - As specified in MIL-STD-202.

4.8.7 Electrical characteristics. When performing electrical tests, the controllers shall be mounted on a suitable heat sink (see 4.4.4).

4.8.7.1 Control signal (see 3.11.1 and 6.5). The control signal shall be verified as specified in 4.8.7.1.1 and 4.8.7.1.2.

4.8.7.1.1 Turn-on signal. With the controller connected as shown in Figure 1, apply rated supply voltage and adjust the load resistance for rated load ±10 percent (see 6.5). Apply the minimum turn-on signal with the control function generator and note that the controller turns ON.

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4.8.7.1.2 Turn-off signal. With the controller ON at rated control signal, apply the maximum turn-off signal with the function generator and note that the controller turns OFF.

4.8.7.2 Turn-on and turn-off time. Measure turn-on and turn-off times with the controller operated as in 4.8.7.1.1 and 4.8.7.1.2.

4.8.7.3 Load voltage rise and fall time. Measure the rise and fall times with the controller operated as in 4.8.7.1 and 4.8.7.1.2.

4.8.7.4 Isolation (see 3.11.4). The power-in terminal, power-out terminal and power-ground terminal shall be shorted together. All remaining terminals shall be shorted together. The points of application shall be the signal ground and power-ground terminals and electrification time shall be two minutes.

4.8.7.4.1 Isolation between controllers (see 3.11.4.1). The power-in, power-out, and ground terminals of each individual controller, all shorted together, shall be shorted with the control, state indication, and signal ground terminals of that same controller. The points of application shall be between the shorted terminals of one controller and the shorted terminals of any other controller on the same printed circuit board. The electrification time shall be two minutes.

4.8.7.5 Control signal (see 3.11.5). With rated supply voltage applied, apply control signal level as specified (see 3.1) and measure control current or voltage. Repeat test for each control signal level specified (see 3.1).

4.8.7.6 Voltage drop. With the controller connected as shown in Figure 1, measure the voltage between the power-in and the power-out terminals while operating at 10, 50 and 100 percent rated load. For ac controllers, a True RMS voltmeter shall be used.

4.8.7.7 Leakage current (see 3.11.7). Connect the controller as shown in Figure 1 with the load resistance adjusted for a maximum of 10 ohms, rated supply voltage applied with the control circuit open, read the leakage current. Repeat the test with the controller command OFF.

4.8.7.8 Power dissipation. Connect the controller as shown in Figure 1 with the load resistance adjusted for short circuit (see 6.5), rated supply voltage applied, and the controller commanded OFF, measure the power dissipation for the OFF state. With the controller commanded ON, measure the power dissipation for the ON state for loads of 10, 50 and 100 percent rated load, unless otherwise specified in the specification sheet.

4.8.7.9 Overload characteristics tests.

4.8.7.9.1 Current limiting (see 3.11.9.1). Connect the controller as shown in Figure 1.

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(a) With the controller off, adjust the load resistor for short circuit (see 6.5) and apply rated supply voltage. While monitoring the current out of the controller, apply rated turn-on signal and measure the peak let-through current (see 6.5) and the current limit level.

(b) With the load resistor adjusted for rated load, apply rated supply voltage and turn-on signal. While monitoring the current out of the controller, apply a short circuit (see 6.5) across the load and measure the peak let-through current (see 6.5) and the current limit level.

4.8.7.9.2 Trip time. Connect the power controller as shown in Figure 1.

(a) Non-repetitive reset (see 6.5) - With rated supply voltage, verify that the controller meets the trip characteristics at various current levels as specified in the specifica- tion sheet.

(b) Repetitive Reset - With rated supply voltage and load resistor adjusted for the specified fault currents, verify that the controller meets the repetitive reset requirements as specified in the specification sheet. Reapply control signal the number of times at the intervals as specified in the specification sheet.

4.8.7.10 Coordination. Connect (with conductors of negligible impedance) the specified (see 3.1) pair of power controllers in series, as shown in Figure 1A. Follow the test procedure of 4.8.7.9.2(a) except that the turn-on signal of the higher rated device shall be applied before the lower rated, and the trip time need not be monitored.

4.8.7.11 Rupture capacity. The controller shall be connected per Figure 1, except that the power source shall be calibrated for the specified rupture current (see 3.1) with the power terminals of the controller shorted. The short shall be removed prior to testing. The open circuit voltage before applica- tion of the rupture current shall be rated voltage (see 3.1). Records of voltage, current and time shall be obtained. The controllers shall be reset within 10 minutes after each rupture test. There shall be sufficient time between rupture tests to allow the power controller temperature to stabilize. The controller shall be subjected to the following test sequence.

(a) Four tests with the controller ON carrying no load current, before application of the rupture condition.

(b) Four tests with the controller OFF, rupture test circuit completed, and rupture condition initiated by the controller being turned on.

(c) After preceding tests, test dielectric withstanding voltage (see 4.8.5.1) and trip time (see 4.8.7.9.2(a)).

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4.8.7.12 Reset function (see 3.11.12). Connect the controller as shown in Figure 1. Apply rated supply voltage and adjust the load resistance for 200 +10% rated load. Adjust control function generator for nominal control signal. Apply the turn-on signal and observe that the controller turns on and trips. Apply the turn-off signal and reapply the turn-on signal in less than the minimum time specified (see 3.1). Observe that the controller does not reset. Apply the turn-off signal and reapply the turn-on signal after the maximum time specified in the specification sheet. Observe that the controller does reset.

4.8.7.13 Deleted.

4.8.7.14 State indication signal(s) (see 3.11.14). Connect the controller as shown in Figure 1. Apply rated voltage and adjust the load resistance for 100% rated load. Apply an OFF command to the control input. Monitor the specified (see 3.1) state indication feedback to verify normal OFF operation and delay times. Apply an ON command to the control input and verify normal ON operation and delay times. Reduce the load current below the specified (see 3.1) minimum current threshold level and verify a fault indication. Increase the load current above the trip threshold and observe that the controller trips and verify specified (see 3.1) trip indication and delay times.

4.8.7.15 Deleted.

4.8.7.16 Ripple voltage (see 3.11.16). With the dc controller connected as shown in Figure 1, monitor the output load voltage and measure the peak-topeak value of ripple voltage generated by the controller under each of the following load conditions: 10% rating and 100% rating (see Figure 4).

4.8.7.17 Trip-free characteristics. With the controller connected as shown in Figure 1, apply rated voltage, adjust load resistor for short circuit and command controller ON. Observe the controller trips out. Reset the controller and command controller ON. Maintain the ON command for one minute minimum and verify that the controller resets and trips only once.

4.8.7.18 Deleted.

4.8.7.19 Operating voltage transients. Connect the controller as shown in Figure 1. Adjust load resistor for rated load.

(a) OFF state. Adjust control function generator for specified (see 3.1) maximum turn-off signal. Apply rated supply voltage for at least 5 seconds and then apply a voltage transient specified (see 3.1). Repeat test for each voltage transient specified (see 3.1).

(b) ON state. For normally OFF controllers. Adjust control function generator for specified (see 3.1) minimum turn-on signal. Apply rated supply voltage for at least 5 seconds and then apply a voltage transient specified (see 3.1). Repeat test for each voltage transient specified (see 3.1).

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4.8.7.20 Transient spike overvoltage. Connect controllers as shown in Figure 1. Adjust load resistor for rated load and control signal for the turn-off signal specified in the specification sheet. Apply spike waveform, as specified below, to the power input terminal of the power controller. The transient shall then be applied to the power output terminal of the controller. The 600 volt dc source shall have (50 ± 10 percent) ohm impedance and shall satisfy the following requirements:

Open-circuit voltage:	+600 volts peak
Rise time:	0.9 ±0.2 microseconds
Fall time:	10.0 ±1.0 microseconds
Pulsewidth (50% amplitude points):	5.0 ±0.1 microseconds
Repetition rate (aperiodic):	Not greater than 50 Hz
Source energy capability:	Not less than 0.01 joule

4.8.7.21 Deleted.

4.8.7.22 Control input transients (see 3.11.21). The following transients shall be applied between the signal ground terminal and the control terminal (source impedance is 250 ohms).

(a) A train of ten pulses of plus and minus 100 volts peak amplitude and 100 microsecond duration each, repeated 10 times at 3-second intervals.

(b) Repeat test (a) between all pairs of active terminals on the unit.

4.8.7.23 Zero voltage turn-on (ZVTO) and zero current turn-off (ZCTO). Connect the controller as shown in Figure 1. Apply rated supply voltage and adjust load impedance for rated load with a 45% lagging power factor. Apply the nominal turn-on signal and subsequently apply the nominal turn-off signal. Monitor the load voltage and current. Repeat test 10 times. Adjust the load impedance to reduce the load current below the specified (see 3.1) minimum and repeat.

4.8.7.24 Deleted.

4.8.7.25 Common mode rejection. With a common mode signal of ±10 volts peak swept from 1.0 Hz to 100 KHz applied between the signal ground and power ground (Figure 1) and with an ON command, verify that the power controller does not turn OFF or does not intermittently operate between ON and OFF. Repeat the test except with an OFF command and insure that the controller does not turn ON.

4.8.7.26 dc offset voltage (ac controllers) (see 3.11.26). With the controller connected as shown in Figure 1, apply rated line voltage and adjust the load impedance to provide current values between 10% and 100% of rating (10% P.F. lagging to 10% leading). With a short circuit applied to the controller from line terminal to load terminal, measure the dc content of the ac supply. Remove the short circuit and measure the dc content at the load terminal. The difference in the two measurements is the dc offset introduced by the controller and shall not exceed the value specified in the specification sheet.

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4.8.7.27 Stabilization time. With the controller connected as shown in Figure 1, the load adjusted for rated current, and supply voltage off, apply an OFF command to the controller. Switch supply voltage to rated voltage and monitor load current. Verify no transient current flow greater than specified in the specification sheet through the load. Remove the supply voltage for sufficient time to stabilize the controller voltages at zero and apply an ON command to the controller. Switch supply voltage to rated voltage and monitor the controller voltage and load current. Verify that the load current flows only when the controller voltage reaches the necessary level within the time interval as specified in the specification sheet and that the current exhibits no switching transients. Interrupt the supply voltage to provide the power outage duration as specified in the specification sheet with the controller still commanded ON. Verify that load current is re-established following the power outage within the time interval specified in the specification sheet.

4.8.8 Fail-safe. The specified number of power controllers must be constructed with their "pass sections" intentionally shorted with no more than 0.05 ohms across any solid state device. Connect one of these shorted power controllers as shown in Figure 1. Adjust load resistors for values of current as specified in the specification sheet. Apply rated load voltage and record time fail-safe element takes to clear. Repeat for each value specified in the specification sheet. If specified (see 3.1) all fail-safe tests shall be conducted with the power controller in an explosive atmosphere as specified in Procedure I, Method 511.2 of MIL-STD-810.

4.8.9 Temperature shock. Controllers shall be tested in accordance with Method 503.2 of MIL-STD-810. The following details and exceptions shall apply:

(a) Special mounting - Controllers, without heat sink, shall be suspended in the test chamber by twine or other non-heat-conducting material in a plane parallel to the normal air flow.

(b) The temperature levels - 71°C, -51°C.

(c) Duration - Four hours each temperature extreme or until temperature stabilizes but not less than one hour.

(d) Number of cycles - Three, each cycle shall be initiated at the high temperature and concluded with the low temperature extreme.

(e) Loading conditions - No load (see 4.4.2).

4.8.10 Operation at temperature extremes (see 3.13). The controller shall be tested for operation at temperature extremes in accordance with the following:

(a) Minimum Temperature - While de-energized, the power controller shall be soaked for 2 hours minimum while mounted on the appropriate heat sink in minimum ambient temperatures specified (see 3.1). At the end of the soak period, while maintaining the specified heat sink temperature, the controller shall meet the requirements of Table V.

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(b) Temperature Range - Apply the rated supply voltage, adjust load to the specified value (see 3.1) and set the control signal to the minimum turn ON value (see 3.1) for normally OFF controllers or the maximum ON value for normally ON controllers. Monitor the output voltage wave form and measure the voltage drop between the power in and power out terminals when the power controller temperature is varied from the minimum specified stabilized temperature (see 3.1) to the maximum specified stabilized temperature (see 3.1).

(c) Maximum Temperature - While de-energized, the power controller shall be soaked for 2 hours minimum while mounted on the appropriate heat sink at the maximum heat sink temperature specified (see 3.1). At the end of the soak period, while maintaining the specified heat sink temperature, the controller shall meet the requirements of Table V.

4.8.11 Vibration (see 3.13). Type 2, 3 and 4 controllers shall be tested in accordance with Procedure I, Method 514 of MIL-STD-810. The following details and exceptions shall apply:

(a) In Step 5 the sinusoidal vibration test curve shown in Figure 2 of this specification shall be used.

(b) The time schedule of Table 514.3-VII shall be used, except the upper sweep frequency shall be 2kHz.

(c) Inspection during test - During the entire vibration schedule, the controller shall be cycled 15 minutes OFF and 15 minutes ON full load (see 6.5). Load voltage and current shall be monitored continuously.

(d) Inspections after tests - As specified in Table V.

Type 1 controllers, when mounted in their intended card cage enclosure, shall be tested to the specified (see 3.1) vibration levels, attitudes, and durations. The following details shall apply:

(a) Inspections during test - During the entire vibration schedule, the controller shall be cycled 15 minutes OFF and 15 minutes ON full load (see 6.5). Load voltage and current shall be monitored continuously.

(b) Inspections after test - As specified in Table V.

4.8.12 Acceleration (see 3.13). Type 1 controllers, mounted in their intended enclosure, and type 2, 3 and 4 controllers shall be tested in accordance with Procedure II, Method 513 of MIL-STD-810. The following details and exceptions shall apply:

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(a) Acceleration level - The highest levels listed for Procedures I and II shall be used.

(b) Test conditions - In each of six directions during Procedure II testing the controller shall be de-energized for one minute with rated supply voltage applied. The controller shall not pass power to the output. In each of six directions, the controller shall be energized for one minute with rated load. The controller shall not show any interruption of output power.

(c) Inspections after test - As specified in Table V.

4.8.13 Salt fog. Controllers shall be tested in accordance with Procedure I, Method 509.2 of MIL-STD-810. The following shall apply:

(a) The duration of the test shall be 48 hours of continuous salt fog exposure, the salt concentration shall be 5±1% and the drying time shall be 48 hours.

(b) Inspections after exposure - Controllers shall be examined for evidence of corrosion, peeling, blistering of the finish, and exposure of base metal.

(c) Inspections after test - After the drying period, the controllers shall be examined as specified in 3.13.1.

4.8.14 Temperature-altitude. Controllers shall be tested in accordance with Procedure III, Method 520.0 of MIL-STD-810. The following details and exceptions shall apply:

(a) The maximum and minimum temperatures shall be as specified (see 3.1). The maximum altitude shall be as specified (see 3.1).

(b) Test item operation - Full load (see 4.4.2).

(c) Heat removal - During controller operation the heat removal apparatus (see 4.4.3) shall be adjusted to allow the case temperature to rise to its highest value but not to exceed the maximum specified (see 3.1). Heat removal parameters shall be monitored to permit the determination of cooling requirements.

4.8.15 Terminal strength. Power controllers shall be tested in accordance with Method 211 of MIL-STD-202 in accordance with the following, as applicable. Unless otherwise specified herein, two terminals of each discrete design, size and configuration shall be tested; however, if there is only one of such design, size and configuration, it shall be tested.

4.8.15.1 Pull test (all terminal types). Terminals shall be tested as specified in test Condition A. The force shall be as specified in the specification sheet (see 3.1).

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4.8.15.2 Bend test (not applicable to plug-in terminals). Terminals shall be tested as specified in test Condition B (two bends) or C, as applicable. Loads for test Condition C shall be as specified (see 3.1).

4.8.15.3 Bend test (plug-in terminals). The lesser of five or all terminals shall be tested in applying a force of five pounds normal to the axis of the pin within 0.125 inch of the tip of the pin.

4.8.15.4 Torque test (screw terminals). All terminals shall be tested as specified in test Condition E as follows:

Screw Size	Torque (Lb-In.)
4-40	4.4
6-32	10.0
8-32	20.0
10-32	32.0
10-24	35.0

Following these tests, power controllers shall be examined for evidence of loosening or breaking of the terminals and other damage that could adversely affect the normal operation of the power controller.

4.8.16 Humidity. Controllers shall be tested in accordance with Procedure III, Method 507.2 of MIL-STD-810. The following details and exceptions shall apply:

(a) Mounting - On a corrosion-resistant panel by normal mounting means.

(b) Inspections during test - During Step 7, the controller shall be subjected to the tests specified in 3.13.1.

NOTE

The dielectric withstanding voltage shall be measured with a test voltage of 90 percent of the initial test potential.

4.8.17 Resistance to soldering heat. Power controllers shall be tested in accordance with Method 210 of MIL-STD-202. The following details and exceptions shall apply:

(a) Depth of immersion in molten solder - Within 0.060 ±.020 inch of the controller base.

(b) Test conditions shall be 232° ±3°C for 60 ±5 seconds.

(c) The inspections after test shall be as specified in 3.12.1.

4.8.18 Electromagnetic compatibility. Controllers shall be tested as specified in MIL-STD-462.

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4.8.19 Shock. Controllers shall be tested in accordance with Procedure IV, Method 516 of MIL-STD-810. The following details and exceptions shall apply:

(a) The pulse configuration shall be half sine wave with level and duration as specified in the specification sheet.

(b) The electrical load conditions - In each of the six directions the controller shall be ON with full load (see 4.4.2) for the first shock pulse and OFF for the second.

(c) The inspections after test shall be as specified in Table V.

4.8.20 Life. Power controllers shall be mounted by their normal mounting means and connected to control maximum rated load at maximum rated output voltage. Controllers shall be continuously cycled ON-OFF at 50 percent duty cycle at 30 cycles per second at maximum rated case temperature for 500 hours, and minimum rated case temperature for 500 hours. Minimum turn-on signal shall be applied to the control input for each ON period and maximum turn-off signal for each OFF period for normally OFF controllers. Use maximum ON signal and minimum OFF signal for normally ON controllers. Once every 24 hours (with a minimum of 12 hours between monitor times), the controller shall be monitored for proper operation for at least 5 minutes. At the end of each temperature period, and with the controller still in the applicable environment, the controller shall meet the requirements of Table V.

4.8.21 Explosive decompression. Subject the controller to an explosive decompression of the surrounding air by changing air pressure from 11.1 psia to 0.65 psia in a time less than 0.02 seconds. The controller shall meet the requirements of Table V.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contract shall be as specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Power controllers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Power controllers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. None required.

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5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), power controllers shall be individually packaged in accordance with the methods of MIL-P-116 designated herein insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Terminals and mounting studs shall be blocked or cushioned to prevent bending or fracture.

5.1.1.4.1 Hermetically-sealed power controllers. Hermetically-sealed controllers shall be individually packaged in accordance with MIL-P-116, Method III.

5.1.1.4.2 Non-hermetically-sealed power controllers. Non-hermetically-sealed controllers shall be individually packaged in accordance with MIL-P-116, IA-8.

5.1.1.5 Intermediate packaging. Power controllers, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube, and shall contain multiples of 5 unit packages, not to exceed 100 packages or 10 pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

5.1.2 Level C. Each cleaned and dried power controller shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged power controllers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirement. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, 2 inches minimum width conforming to PPP-T-60, Class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636, using nonmetallic or tape banding only.

5.2.2 Level B. The packaged power controllers shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirement. Closures shall be in accordance with the appendix thereto. For Army procurement, fiberboard containers shall be class weather-resistant as specified in Level A.

5.2.3 Level C. The packaged power controllers shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. This pack shall conform to the applicable carrier rules and regulations.

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5.2.4 Unitized loads. Unitized loads, commensurate with the level of packaging specified in the contract or order, shall be used whenever quantities for shipment to one destination equal 50 cubic feet or more.

5.2.4.1 Level A. Power controllers, packed as specified in 5.2.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap, having two sides extending down the stacked load at least 12 inches, shall be positioned over the load to accommodate marking requirements. The cap shall be in accordance with PPP-F-320, class weather resistant. The load shall be strapped to the pallet.

5.2.4.2 Level B. Power controllers, packed as specified in 5.2.2, shall be unitized as specified in 5.4.2.1, except that fiberboard cap shall be class domestic. For Army use only, caps shall be weather resistant as specified in Level A.

5.2.4.3 Level C. Power controllers, packed as specified in 5.2.3, shall be unitized as specified in 5.2.4.1, except that pallet and cap shall be of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible.

5.5 Inspection. Inspection of military packaging shall be in accordance with 4.7.3.

6. NOTES

6.1 Intended use. Power controllers conforming to this specification are intended for use in controlling the making and breaking of power circuits for electrically operated equipment and devices; and for providing overload and short-circuit protection in direct and alternating current (single or multi-phase) electrical systems. Their principal areas of application are for aircraft, missiles, spacecraft and ground support equipment. This does not preclude the use of power controllers in other military applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Title, number and date of the applicable specification sheet and the military part number.

(c) The packaging and packing level required (see Section 5).

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6.2.2 Qualified controllers with modification (see 3.2.2).

(a) Title, number and date of this specification.

(b) Title, number and date of applicable specification sheet for similar power controller.

(c) Military part number of similar qualified power controller.

(d) Manufacturer's part number of modified power controller.

(e) Details of the variations from the specification sheet.

(f) Inspection requirements.

(1) Tests to be performed (if any).

(2) The laboratory at which inspection is to be performed.

(3) Samples and submission data, if other than that specified.

(g) The packaging and packing level specified (see Section 5).

6.3 Indirect shipments. The preservation and packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products Lists whether or not such products have actually been so listed by this date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Washington, DC 20361, and information pertaining to qualification of products may be obtained from that activity.

6.4.1 Copies of "Provisions Government Qualification SD-6" may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Definitions. The definitions listed below are not a complete glossary of solid state power controller terminology, but rather are intended as definitions of the technical terms as applied within this specification.

6.5.1 Power controller. A power controller is a device providing an output power switch which presents a low impedance between its supply and load terminals when in the ON state and a high impedance in the OFF state.

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The state of the power switch normally conforms to that represented by a signal externally applied to the controller.

The state of the power switch is represented by an indication signal.

The controller trips to isolate the load circuit from the supply on detection of an electrical overload, or limits the current to a specified value.

6.5.2 INTENTIONALLY BLANK.

6.5.3 INTENTIONALLY BLANK.

6.5.4 Current limiting. A protective characteristic of a controller which limits the output current to a specified value. This limit thus defines the maximum value of overload current prior to tripping.

6.5.5 Trip. The automatic interruption of current which results from electrical overloads.

6.5.6 Trip-free. A characteristic which will not allow a device to be maintained ON with a continued application of an input signal when carrying an overload current that would normally cause the device to trip.

6.5.7 Zero voltage turn-on (ZVTO) and zero current turn-off (ZCTO). A characteristic that requires the power controller to turn ON and turn OFF only at the half-cycle zero crossing point when connected to a resistive load, regardless of when the control signal is applied or removed.

6.5.8 Trip time. The time lapse between the application of an overload or fault and the 10 percent value of rated output current.

6.5.9 Reset. The restoration of the tripped power controller to a state from which it can be turned ON.

6.5.10 Turn-on signal. The control signal level at which the power controller is turned ON.

6.5.11 Turn-off signal. The control signal level at which the power controller is turned OFF.

6.5.12 Supply voltage. The voltage applied between the power input terminal of the controller and the power ground.

6.5.13 Load voltage. The voltage between the power output terminal of the controller and the power ground.

6.5.14 Turn-on time. See Figure 3.

6.5.15 Turn-off time. See Figure 3.

6.5.16 Rise time. See (a) of Figure 3.

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6.5.17 Fall time. See (a) of Figure 3.

6.5.18 INTENTIONALLY BLANK.

6.5.19 ON state. That condition which, with the turn-on signal applied, the device allows power to be passed to the load.

6.5.20 OFF state. That condition which, with the turn-off signal applied, the device prevents power from being passed to the load.

6.5.21 Short circuit. A short circuit is defined as a very low impedance applied between the output terminal and ground.

6.5.22 Application time to reset. The time interval the reset signal must be applied to cause reset (see 3.1).

6.5.23 Removal time to reset. The time interval the control signal must be removed before reapplication to cause reset (see 3.1).

6.5.24 Nonrepetitive reset. The minimum specified (see 3.1) time interval between tripout and reset for which the trip characteristics are valid.

6.5.25 Rupture current. The value of current in a circuit which reflects the capabilities of the power source without the effects of the controller.

6.5.26 Power dissipation. When the power controller is ON, the power dissipation shall include all power dissipated in the power switching circuit, power losses due to internal leakage currents, and power supplies. Control input and reset signals shall be excluded. When power controller is OFF, the power dissipation shall include only dissipation due to leakage currents and internal power supplies.

6.5.27 Peak let through current. The maximum value of current available to the load through the device during the first 100 microseconds after application of the overload or fault condition (see Figure 4).

6.6 Drawing preparation. As an aid in the preparation of drawings the following power controller characteristics (when applicable) should be considered.

Paragraph No.	Characteristic
3.5	Design
	Weight
	Physical Dimension
3.5.3	Mounting Hardware
3.5.4.1	Mounting Bracket
3.5.5.1.1	Terminal Location
3.11.1	Control Signal Turn-On
	Control Signal Turn-Off

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Paragraph No.	Characteristic
3.11.2	Turn-on time
	Turn-off time
3.11.3	Load voltage rise time
	Load voltage fall time
3.11.4	Isolation Voltage
3.11.6	Voltage Prop
3.11.7	Leakage Current
3.11.8	Power Dissipation
3.11.9.1	Current Limiting
3.11.9.2	Trip time
3.11.10	Coordination
3.11.12	Reset signal
3.11.13.1	Minimum time to reset
	Maximum time to reset
3.11.13.2	Removal time to reset
	Minimum
	Maximum
3.11.14	State indication
3.11.16	Ripple current (dc controller)

6.7 Subject term (keyword) listing.

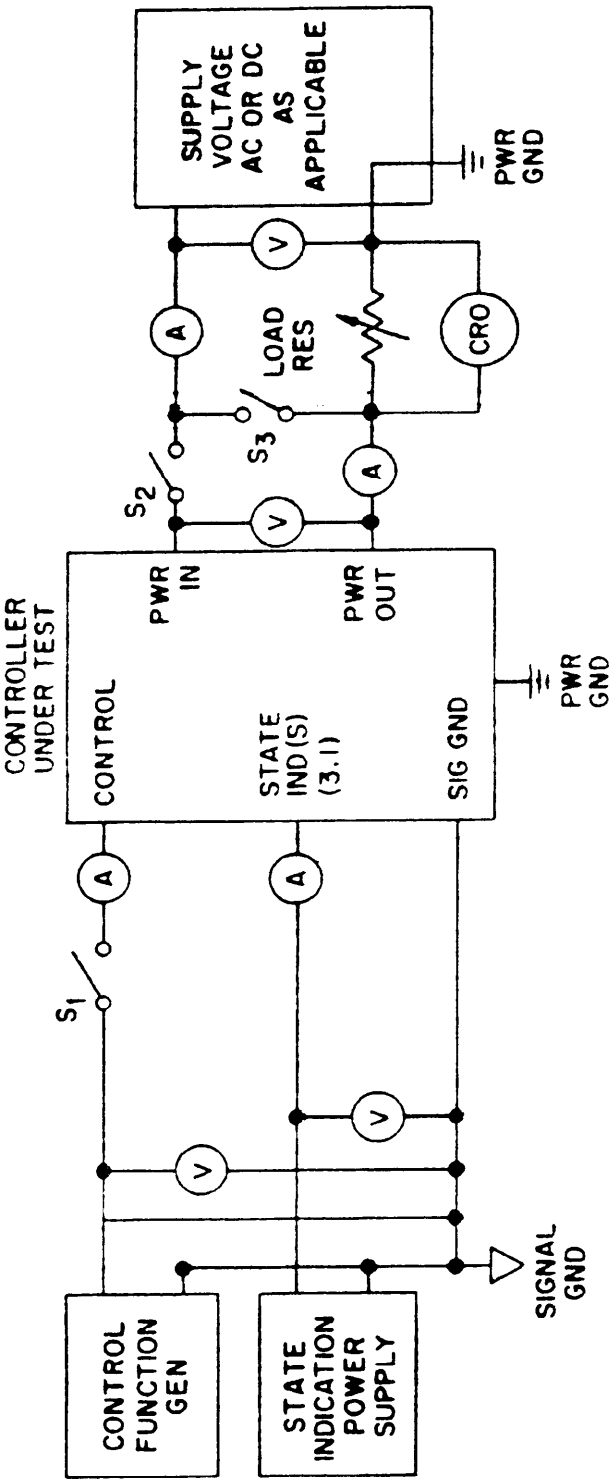
ac power controller
dc power controller
Power controller
Solid state power controller

Custodians:
Navy - AS
Air Force - 11
Army - ME

Review activity:
Air Force - 99

Preparing activity:
Navy - AS
(Project No. 6110-N293)

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NOTE: S_1 IS A BOUNCELESS SWITCH.

FIGURE 1. Test circuit (4.8.7, 4.8.8).

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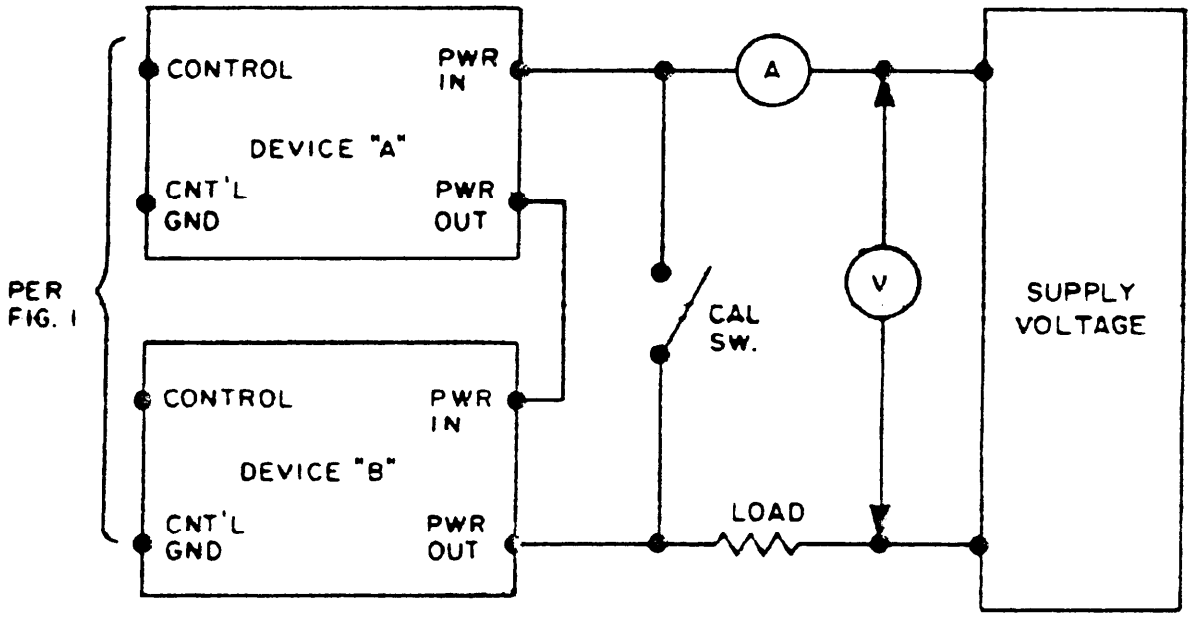


FIGURE 1A. Coordination test circuit (4.8.7.10).

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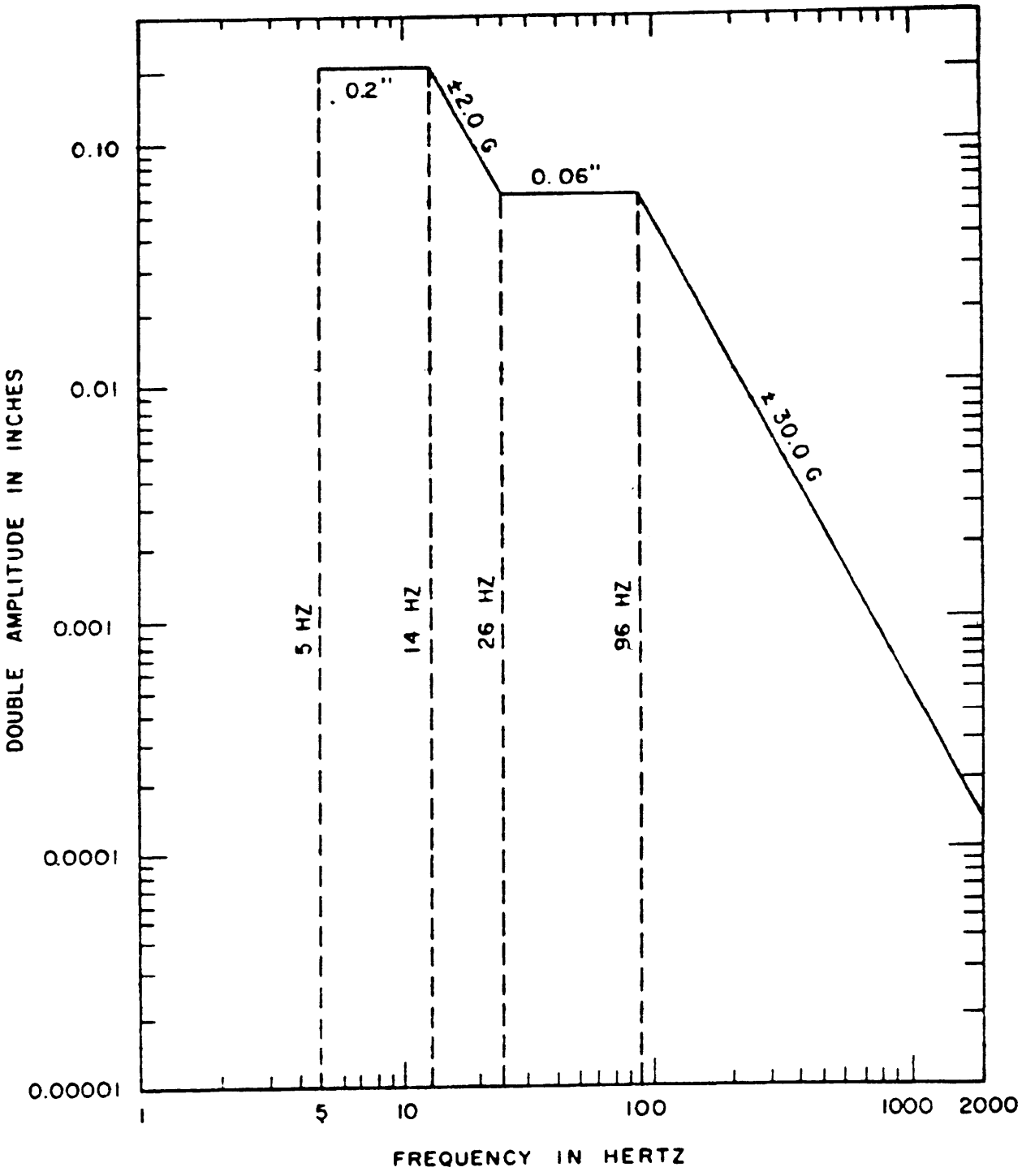


FIGURE 2. Vibration level curve (4.8.11).

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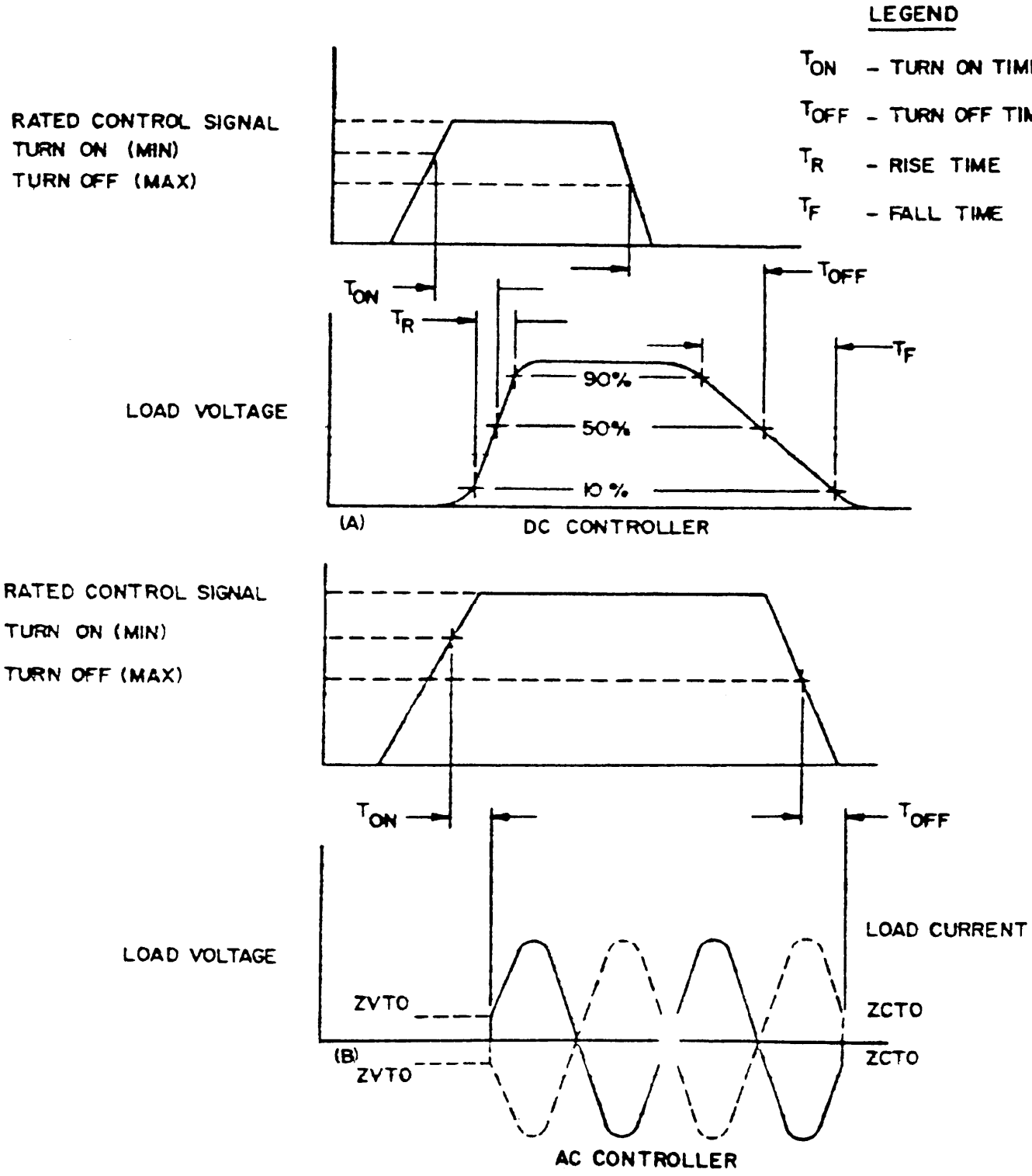


FIGURE 3. Illustration of timing characteristics. (6.5.14, 6.5.15, 6.5.16, 6.5.17)

NOTE: (1) Normally off controller as shown; reverse on/off designations for normally on controller.

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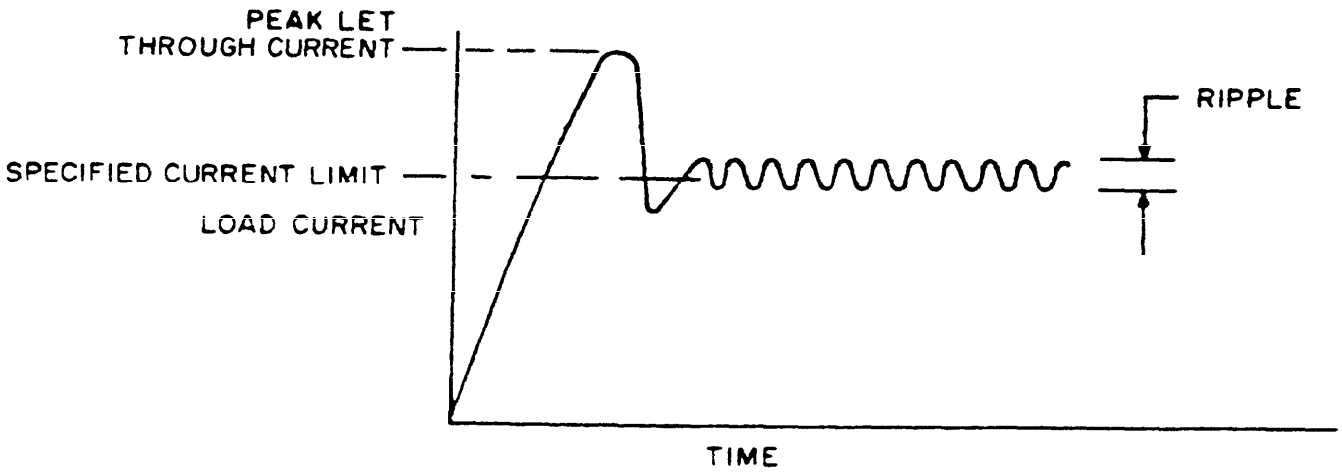


FIGURE 4. Overload let through current (6.5.27).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-0081653C(AS)	POWER CONTROLLER, SOLID STATE, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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